

From the Bilocal G – Σ Action to the Coadjoint-Orbit Bosonization Action: A Step-by-Step Reduction and Its Status

Status: plausible.

This document does *not* present a complete rigorous derivation. The reduction of the bilocal G – Σ action to the coadjoint-orbit action *closes in form*: the algebra that manufactures the Berry/Wess–Zumino and Hamiltonian terms is exact and independently checked. However, the bridge rests on **two uncontrolled steps**, each made physically plausible but neither supplied with a controlled higher-order bound:

- (P1) **Elimination of the self-energy Σ by its Dyson saddle.** The relevant integral is genuinely non-Gaussian (it carries a $\det$ factor); a one-mode model shows the saddle returns the wrong leading dependence on G . It is rendered controlled only by introducing N identical flavors, a parameter that is *fictitious* for the genuine single-species Fermi gas. For one species the step is only plausible.
- (P2) **Restriction of the bilocal G to the projector orbit.** The discarded transverse “amplitude” fluctuations lie in the *same gapless* particle–hole continuum as the orbit modes; no spectral gap protects the truncation. The suppression is a semiclassical phase-space-measure (Haldane-patch) argument in $q/k_F \ll 1$, made plausible but not bounded uniformly.

Moreover, joint control of (P1) and (P2) by a single physical parameter q/k_F is asserted, not proved. An *exact* bridge is structurally foreclosed: any exact integration of the starting action trivializes it (see Step 1).

1 The two endpoints, with all conventions inlined

1.1 Starting point: the bilocal G – Σ action

We work with spinless nonrelativistic fermions in d spatial dimensions at inverse temperature β , in the imaginary-time coherent-state path integral. A collective coordinate $1 \equiv (\mathbf{x}_1, \tau_1)$ is used, with

$$\int d1 := \int_0^\beta d\tau_1 \int d^d x_1, \quad \delta(1, 2) := \delta(\tau_1 - \tau_2) \delta^d(\mathbf{x}_1 - \mathbf{x}_2).$$

The single-particle Hamiltonian is $\hat{h} = \varepsilon(-i\nabla) - \mu$ with $\xi_{\mathbf{k}} := \varepsilon(\mathbf{k}) - \mu$. Traces of bilocal kernels are $\text{Tr}(AB) := \int d1 d2 A(1, 2)B(2, 1)$, and $\text{Tr} \ln$ is defined through its series. The free inverse propagator is the operator with kernel

$$G_0^{-1}(1, 2) := -(\partial_{\tau_1} + \hat{h}_1) \delta(1, 2), \quad G_0(\mathbf{k}, i\omega_n) = \frac{1}{i\omega_n - \xi_{\mathbf{k}}},$$

on antiperiodic Matsubara frequencies $\omega_n = (2n + 1)\pi/\beta$.

Introducing the bosonic bilocal $G(1, 2)$ together with a conjugate self-energy field $\Sigma(1, 2)$ enforcing $G(1, 2) = -\psi(1)\bar{\psi}(2)$, and integrating out the Grassmann fields exactly, yields the exact bosonic action

$$S[G, \Sigma] = -\text{Tr} \ln(G_0^{-1} + \Sigma) + \text{Tr}(\Sigma G), \quad \text{Tr}(\Sigma G) := \int d1 d2 \Sigma(2, 1) G(1, 2), \quad (1)$$

(the free case; any interaction adds $S_{\text{int}}[G]$, set to zero here). Its saddle equations are the Dyson equation and the self-energy condition,

$$G = (G_0^{-1} + \Sigma)^{-1}, \quad \Sigma = -\frac{\delta S_{\text{int}}}{\delta G},$$

so that for the free theory $\Sigma_\star = 0$, $G_\star = G_0$. The content lies in the fluctuations of G about G_0 . At zero temperature and equal times the free propagator reduces to the occupation $n_{\mathbf{k}} \in \{0, 1\}$, i.e. the single-particle operator $\hat{n}$ is a projector, $\hat{n}^2 = \hat{n}$.

1.2 Proposed endpoint: the coadjoint-orbit action

Single-particle observables are functions $F(\mathbf{x}, \mathbf{p})$ on phase space. Operator multiplication maps to the star product

$$F \star G = F \exp\left(\frac{i\hbar}{2}(\overleftarrow{\partial}_{\mathbf{x}} \overrightarrow{\partial}_{\mathbf{p}} - \overleftarrow{\partial}_{\mathbf{p}} \overrightarrow{\partial}_{\mathbf{x}})\right) G, \quad (2)$$

and commutators to the Moyal bracket

$$\{F, G\}_{\text{M}} = -i(F \star G - G \star F) = 2F \sin\left(\frac{\hbar}{2}(\overleftarrow{\partial}_{\mathbf{x}} \overrightarrow{\partial}_{\mathbf{p}} - \overleftarrow{\partial}_{\mathbf{p}} \overrightarrow{\partial}_{\mathbf{x}})\right) G.$$

Distribution functions f are paired with observables through

$$\langle f, F \rangle := \int \frac{d^d x d^d p}{(2\pi)^d} f(\mathbf{x}, \mathbf{p}) F(\mathbf{x}, \mathbf{p}).$$

In the semiclassical regime $\hbar \nabla_{\mathbf{x}} \cdot \nabla_{\mathbf{p}} \ll 1$ the star product reduces to ordinary multiplication and the Moyal bracket to the Poisson bracket $\{F, G\} = \partial_{\mathbf{x}} F \cdot \partial_{\mathbf{p}} G - \partial_{\mathbf{p}} F \cdot \partial_{\mathbf{x}} G$. We set $\hbar = 1$ except where shown.

Canonical transformations $U = \exp(i\phi)$ act on a distribution by conjugation, $f = U \star f_0 \star U^{-1}$, the coadjoint action through the Fermi-sea base point

$$f_0(\mathbf{x}, \mathbf{p}) = \Theta(p_F - |\mathbf{p}|),$$

the Wigner image of the equal-time projector $\hat{n}$. The proposed bosonized action of the noninteracting Fermi surface is

$$S[f] = S_{\text{WZW}}[f] + S_H[f], \quad S_{\text{WZW}} = \int dt \langle f_0, U^{-1} \star i \partial_t U \rangle, \quad S_H = - \int dt \langle f, \epsilon(\mathbf{p}) \rangle, \quad (3)$$

with $\epsilon(\mathbf{p}) = \varepsilon(\mathbf{p}) - \mu$ the $\xi_{\mathbf{k}}$ above. S_{WZW} is the path integral of the Kirillov–Kostant–Souriau symplectic form on the orbit; its equation of motion is the collisionless quantum Boltzmann equation $\partial_t f = \{\epsilon, f\}_{\text{M}}$.

Claim under test. That (1) reduces directly to (3). We now carry the reduction step by step, labeling each step *exact*, *controlled* (with named parameter and what is dropped), or *plausible* (with the supporting argument), and stopping wherever a step ceases to be controlled.

2 The reduction

Step 0 — The starting action (exact)

Equation (1) is the exact bilocal action of the free theory; it is the object we transform. **Status: exact.**

Step 1 — Why no exact integration suffices (exact, structural)

In (1) the bilocal G appears *only linearly*, through $\text{Tr}(\Sigma G)$. Integrating G first is therefore exact and produces a functional delta:

$$\int \mathcal{D}G e^{-\text{Tr}(\Sigma G)} \propto \prod_{1,2} \delta(\Sigma(2,1)) \implies \Sigma = 0, \quad Z = e^{\text{Tr} \ln G_0^{-1}} = Z_{\text{free}}.$$

Status: exact, and structurally decisive: *any* exact elimination of either field returns the free partition function with no orbit action. The coadjoint action can therefore arise only from a *saddle* in Σ together with a *restriction* of the domain of G . An exact bridge is foreclosed; the best attainable is a controlled saddle-plus-truncation.

Step 2 — Eliminating Σ (P1; uncontrolled for one species)

Shift $M := G_0^{-1} + \Sigma$ (Jacobian unity, exact). The Σ -integral becomes

$$\int \mathcal{D}\Sigma e^{-S} = e^{\text{Tr}(G_0^{-1}G)} I[G], \quad I[G] = \int \mathcal{D}M e^{\text{Tr} \ln M - \text{Tr}(MG)}.$$

The integrand carries the factor $\text{Tr} \ln M$ (equivalently $\det M$): $I[G]$ is *not Gaussian* and *not* a linear Lagrange-multiplier integral. Its stationary point is the Dyson equation $M_\star = G^{-1}$, which gives the two-particle-irreducible functional

$$S_{\text{eff}}[G] = \text{Tr} \ln G - \text{Tr}(G_0^{-1}G) + \text{const.} \quad (4)$$

The fluctuations of M about G^{-1} are not negligible for a single species. A one-mode caricature ($\text{Tr} \rightarrow \text{scalar}$) is exact and damning:

$$\int_0^\infty dm e^{\ln m - mg} = \int_0^\infty dm m e^{-mg} = g^{-2}, \quad \text{whereas the saddle gives } e^{\ln g^{-1} - 1} = g^{-1} e^{-1}.$$

The fluctuations change the leading power $g^{-1} \rightarrow g^{-2}$; the saddle is therefore *uncontrolled as written*.

Controlled rescue (a different model). With N identical flavors and a flavor-singlet bilocal, $S \rightarrow NS$ and $I[G] = \int \mathcal{D}M e^{N(\text{Tr} \ln M - \text{Tr} MG)}$. The caricature becomes

$$\int_0^\infty dm e^{N(\ln m - mg)} = \frac{\Gamma(N+1)}{(Ng)^{N+1}} = \exp \left[\underbrace{-N \ln g - N}_{O(N) \text{ saddle}} + O(\ln N) \right],$$

so the saddle reproduces the $O(N)$ action with remainder $O(\ln N/N)$. **Status: controlled for the N -flavor model** (small parameter $1/N$, dropping $O(\ln N/N)$), but **plausible at best for the genuine single-species Fermi gas**, where no such N exists. The physical hope is that large local occupation per phase-space patch (Step 5) plays the role of large N ; this is not established for the self-energy sector. We carry (4) forward, flagging it as the weakest link.

Step 3 — Restriction to the projector orbit (P2; a restriction, not an identity)

We impose the ansatz that the low-energy configurations are images of the Fermi-sea projector under canonical transformations. As a two-time kernel,

$$G = \hat{U} G_0 \hat{U}^{-1}, \quad \hat{U} = \hat{U}(\tau) = e^{i\hat{\phi}(\tau)} \quad (\text{single-particle, time-local}), \quad (5)$$

whose equal-time projection is $\hat{U}(\tau)\hat{n}\hat{U}^{-1}(\tau) = \hat{f}(\tau)$, matching $f = U \star f_0 \star U^{-1}$. This is a *restriction of the integration domain*, not an identity. A general fluctuation δG about G_0 splits into orbit directions $\delta G_{\parallel} = [i\hat{\phi}, G_0]$ and transverse “amplitude” directions $\delta G_{\perp}$ that move occupation eigenvalues off $\{0, 1\}$. The quadratic action from (4),

$$\delta^2 S_{\text{eff}} = -\frac{1}{2} \text{Tr}(G_0^{-1} \delta G G_0^{-1} \delta G),$$

is the inverse particle–hole propagator — the linearized Lindhard/RPA kernel — and *both* $\delta G_{\parallel}$ and $\delta G_{\perp}$ inhabit the *same gapless* particle–hole continuum. No spectral gap separates the orbit from its complement. The status of this restriction is assessed in Step 5; the algebra it enables (Step 4) is exact.

Step 4 — Exact reduction of S_{eff} on the orbit (exact)

Write $D := \partial_{\tau} + \hat{h}$, so $G_0^{-1} = -D$ and $G_0 = -D^{-1}$.

(4a) Entropy term. By multiplicativity of $\ln \det$ under similarity,

$$\text{Tr} \ln(\hat{U} G_0 \hat{U}^{-1}) = \text{Tr} \ln G_0.$$

Status: exact for invertible time-local $\hat{U}$. (For τ -dependent $\hat{U}$ a regularization-dependent anomaly in $\text{Tr} \ln$ can shift only the *normalization* of the Berry term, never its form.) An independent finite-dimensional evaluation confirms $\text{Tr} \ln G = \text{Tr} \ln G_0$ to machine precision (residual $\sim 10^{-11}$, pure floating-point roundoff in the log-determinant). The entropy term is thus inert on the orbit and carries no dependence on $\hat{U}$.

(4b) The kinetic/Hamiltonian term. With the exact operator identity $D\hat{U} = \hat{U}D + [D, \hat{U}]$ and $[D, \hat{U}] = (\partial_{\tau}\hat{U}) + [\hat{h}, \hat{U}]$,

$$D\hat{U}D^{-1}\hat{U}^{-1} = 1 + [(\partial_{\tau}\hat{U}) + [\hat{h}, \hat{U}]]D^{-1}\hat{U}^{-1}. \quad (6)$$

Since $-\text{Tr}(G_0^{-1}G) = -\text{Tr}(D\hat{U}D^{-1}\hat{U}^{-1})$ and $D^{-1} = -G_0$,

$$-\text{Tr}(G_0^{-1}G) = -\text{Tr}(1) + \text{Tr}((\partial_{\tau}\hat{U})G_0\hat{U}^{-1}) + \text{Tr}([\hat{h}, \hat{U}]G_0\hat{U}^{-1}). \quad (7)$$

Equations (6)–(7) are exact operator algebra; an independent finite-dimensional check finds the identity (7) to hold as a matrix identity (difference = 0 exactly). **Status: exact.**

(4c) Equal-time collapse. Each surviving trace has G_0 sandwiched between time-local factors, so $G_0(\tau, \tau) \rightarrow \hat{f}_0 = \hat{n}$ under the standard 0^+ point-splitting. By cyclicity,

$$\text{Tr}((\partial_{\tau}\hat{U})G_0\hat{U}^{-1}) = \int_0^{\beta} d\tau \text{tr}(\hat{U}^{-1}(\partial_{\tau}\hat{U})\hat{f}_0),$$

$$\text{Tr}([\hat{h}, \hat{U}]G_0\hat{U}^{-1}) = \int_0^\beta d\tau [\text{tr}(\hat{h}\hat{f}) - \text{tr}(\hat{h}\hat{f}_0)], \quad \hat{f} = \hat{U}\hat{f}_0\hat{U}^{-1}.$$

Status: exact up to the equal-time prescription (the point-splitting / 0^+ convention), which can affect only the Berry-term normalization. Collecting and dropping τ -independent constants,

$$S_{\text{eff}}|_{\text{orbit}} = \int_0^\beta d\tau \text{tr}(\hat{f}_0 \hat{U}^{-1} \partial_\tau \hat{U}) + \int_0^\beta d\tau \text{tr}(\hat{h} \hat{f}) + \text{const.} \quad (8)$$

(4d) Wick rotation. With $\tau = it$, $\int d\tau = i \int dt$, $\partial_\tau = -i\partial_t$, and $-S_E = iS_{\text{real}}$,

$$\boxed{S_{\text{real}} = \int dt \text{tr}(\hat{f}_0 \hat{U}^{-1} i\partial_t \hat{U}) - \int dt \text{tr}(\hat{f} \hat{h}), \quad \hat{f} = \hat{U} \hat{f}_0 \hat{U}^{-1}.} \quad (9)$$

This is the first-quantized form of the proposed action. The Berry/Wess–Zumino term descends from the ∂_τ in G_0^{-1} (the Kirillov–Kostant–Souriau symplectic structure on the orbit through $\hat{f}_0$), and the Hamiltonian term from $\hat{h}$. **Status: exact**, modulo the (4a)/(4c) regularization caveats.

Step 5 — Wigner/Moyal map, and the honest status of the truncation

(5a) Wigner transform (exact). The Wigner transform

$$\omega(\mathbf{x}, \mathbf{p}) = \int d^d y \, e^{i\mathbf{p} \cdot \mathbf{y}/\hbar} \langle \mathbf{x} - \frac{\mathbf{y}}{2} | \hat{\omega} | \mathbf{x} + \frac{\mathbf{y}}{2} \rangle$$

is an exact operator $\leftrightarrow$ symbol isomorphism with $\text{Tr}(\hat{a}\hat{b}) = \int \frac{d^d x d^d p}{(2\pi\hbar)^d} a \star b$ and the symbol of a product equal to the star product (2). Applying it term by term to (9) gives, with $f = U \star f_0 \star U^{-1}$ and $\int f \star \epsilon = \int f \epsilon$,

$$\boxed{S = \int dt \int \frac{d^d x d^d p}{(2\pi\hbar)^d} \left[f_0 (U^{-1} \star i\partial_t U) - f \epsilon(\mathbf{p}) \right] = S_{\text{WZW}} + S_H.} \quad (10)$$

Status: exact. The transform retains the *full Moyal star*, not its Poisson truncation, so this step requires no $\hbar \rightarrow 0$; the nonlinear Moyal structure of (3) is reproduced intact. This is exactly the proposed coadjoint-orbit action, with f on the orbit through $f_0 = \Theta(p_F - |\mathbf{p}|)$.

(5b) Status of the orbit restriction (P2; plausible, not controlled). Steps 4–5a manufacture the target action exactly *given* the ansatz (5); the remaining question is the legitimacy of discarding $\delta G_\perp$. As noted in Step 3, these modes are not gapped away. The only available suppression is semiclassical/Haldane-patch counting: a coherent deformation of wavevector $q \ll k_F$ acts collectively on $N_{\text{patch}} \sim (k_F/q)^{d-1}$ fermions, whereas an amplitude fluctuation is a single particle–hole excitation carrying relative weight $\sim 1/N_{\text{patch}} \sim (q/k_F)^{d-1}$. This identifies the physical small parameter as $\hbar \nabla_{\mathbf{x}} \cdot \nabla_{\mathbf{p}} \sim q/k_F \ll 1$ and matches the counting that controls Fermi-surface bosonization. It is a concrete, literature-backed argument, and it correctly reproduces the leading-order density response. *But* because the modes are gapless, the suppression is a phase-space-measure effect, not an energetic gap, and no uniform higher-order bound on the $\delta G_\perp$ contribution to the action is supplied here. **Status: plausible, not controlled.**

(5c) Joint control (asserted). The parameter $1/N$ that made Step 2 controlled is fictitious for the single-species gas; the honest expectation is that the single physical parameter q/k_F (large patch occupation) controls both the Σ -saddle (P1) and the off-orbit truncation (P2). A concrete patch-counting mechanism is given only for the orbit sector. That the same parameter bounds the self-energy fluctuations is **asserted, not proved**.

3 Summary of the chain

Step	Content	Status
0	starting bilocal action (1)	exact
1	exact integration trivializes $\Rightarrow$ need saddle+restriction	exact (structural)
2	eliminate $\Sigma \Rightarrow S_{\text{eff}} = \text{Tr} \ln G - \text{Tr}(G_0^{-1}G)$	P1: controlled only with fictitious $1/N$; else p
3	restrict G to projector orbit (5)	restriction (off-orbit modes gapless)
4	reduce on orbit to first-quantized (9)	exact (minor regularization caveats)
5a	Wigner transform $\rightarrow$ Moyal action (10)	exact (full Moyal, no $\hbar \rightarrow 0$)
5b	justify dropping $\delta G_{\perp}$	P2: plausible ($q/k_F \ll 1$), not controlled
5c	joint control of P1 and P2 by one parameter	asserted, not proved

The term-manufacturing algebra (Steps 4–5a) is exact: the Berry/Wess–Zumino term is the Kirillov–Kostant–Souriau form descending from ∂_{τ} in G_0^{-1} , the Hamiltonian term descends from $\hat{h}$, and the full Moyal structure survives. The difficulty does *not* live there. It lives entirely in the two eliminations: eliminating Σ (P1) and restricting G to the orbit (P2).

4 Caveats / status of approximations

- **No exact bridge.** Step 1 shows that any exact integration of (1) returns Z_{free} with no orbit action. The coadjoint-orbit action exists only as a *saddle-plus-truncation*; “exact” is structurally foreclosed.
- **(P1) The Σ -elimination is the weakest link.** The integral $I[G] = \int \mathcal{D}M e^{\text{Tr} \ln M - \text{Tr}(MG)}$ is non-Gaussian; the one-mode model shows the Dyson saddle returns the wrong leading G -dependence (g^{-1} vs. exact g^{-2}). The N -flavor computation controls a *different* theory ($O(\ln N/N)$ remainder) and does not transfer to one species. To promote this step one would need, in the single-species model, an explicit large prefactor $\lambda \gg 1$ with the Σ -fluctuation correction bounded as $O(1/\lambda)$ — concretely, the Gaussian fluctuation determinant of δM about G^{-1} *on the orbit*, shown to be higher order in q/k_F . This is not done here.
- **(P2) The orbit restriction is the fundamental obstruction.** The transverse amplitude modes share the gapless particle–hole continuum with the orbit modes; their suppression is a phase-space-measure argument ($q/k_F \ll 1$, $N_{\text{patch}} \gg 1$), not an energetic gap. To promote it one would need a uniform bound that the $\delta G_{\perp}$ contribution to the action is $O((q/k_F)^{\#})$ relative to the orbit terms; being a measure effect, not a gap, this is not supplied.
- **Joint control unestablished.** Promoting both P1 and P2 to controlled requires a

single physical parameter q/k_F to bound the self-energy sector and the off-orbit sector simultaneously, with explicit higher-order estimates. Only the orbit sector is argued concretely; the self-energy sector is, by the present reasoning, merely expected to follow.

- **Regularization.** The Tr ln is UV-divergent; only the difference $\text{Tr ln}(G_0^{-1} + \Sigma) - \text{Tr ln } G_0^{-1}$ is finite, and equal-time objects (4c) require point-splitting / a $e^{i\omega_n 0^+}$ convergence factor. These prescriptions can shift the Wess–Zumino *normalization* but not the form of (10).
- **Measure and contour.** The bosonic measure $\mathcal{D}G \mathcal{D}\Sigma$, the imaginary contour for Σ , and the Jacobian from $\psi, \bar{\psi}$ to G are treated formally; they do not affect the saddle structure but matter for fluctuation determinants (hence for any attempt to control P1).
- **Real vs. imaginary time.** The starting action is in imaginary time; the target is in real time. The Wick rotation (4d) aligns the ∂_τ of the Tr ln sector with the $i\partial_t$ of the Berry term; subtleties of contour and of total-derivative/ θ -terms in the Wess–Zumino term in compact momentum directions are not analyzed here.
- **Moyal vs. Poisson.** The Wigner map (5a) is exact and keeps the full Moyal algebra; the Poisson form is its semiclassical truncation $\hbar \nabla_{\mathbf{x}} \cdot \nabla_{\mathbf{p}} \ll 1$. This distinction is physical and is preserved by the present route.

Overall. The reduction *closes in form*: the bilocal G – Σ action yields the coadjoint-orbit action (10) once Σ is eliminated by its saddle and G is restricted to the projector orbit. Both eliminations are supported by concrete physical arguments (large patch occupation / semiclassical q/k_F , with the N -flavor computation as partial evidence for the saddle), but neither is supplied with a controlled higher-order bound for the single-species Fermi gas, and their joint control is not proved. The derivation is therefore *plausible*, not controlled or exact.